

1.4 Solving Quadratic Equations

Definition

Clarifying Parentheses, Brackets, and Braces:

- $()$ — Used for **ordered pairs** or **open intervals**, where endpoints are not included. Example: $(2, 5)$ represents all numbers between 2 and 5, not including 2 or 5.
- $[]$ — Used for **closed intervals**, where endpoints are included. Example: $[2, 5]$ includes 2 and 5.
- $\{\}$ — Used to list **sets** or **solution sets**. Example: $\{-3, 3\}$ means the equation has two solutions, -3 and 3 .

Definition

Quadratic Equation:

A **quadratic equation** is any equation that can be written in the form

$$ax^2 + bx + c = 0$$

where a, b , and c are real numbers and $a \neq 0$. The highest power of x is 2, making it a second-degree equation.

Definition

Zero Product Property:

If $mn = 0$, then $m = 0$ or $n = 0$.

This allows us to solve factored quadratic equations by setting each factor equal to zero.

Example: Solve by factoring.

$$x^2 - 7x = 0$$

Solution:

Factor: $x(x - 7) = 0$

Set each factor equal to zero: $x = 0$ or $x = 7$

Solution set: $\{0, 7\}$.

Definition

Square Root Property:

If $x^2 = k$, then $x = \pm\sqrt{k}$, provided $k \geq 0$.

Example: Solve using the square root property.

$$(x - 4)^2 = 9$$

Solution:

Take the square root of both sides:

$$x - 4 = \pm 3$$

Add 4:

$$x = 4 \pm 3$$

Solution set: $\{1, 7\}$.

Definition

Perfect Square Trinomials:

A **perfect square trinomial** is a quadratic expression that can be written as the square of a binomial.

Examples:

$$(x + 3)^2 = x^2 + 6x + 9, \quad (x - 2)^2 = x^2 - 4x + 4.$$

In general:

$$(x + n)^2 = x^2 + 2nx + n^2.$$

Definition

Completing the Square:

To solve a quadratic equation of the form $x^2 + bx + c = 0$, we can create a perfect square trinomial by adding $\left(\frac{b}{2}\right)^2$ to both sides.

$$x^2 + bx = -c \quad \Rightarrow \quad x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(\frac{b}{2}\right)^2 - c.$$

Then factor the left-hand side:

$$\left(x + \frac{b}{2}\right)^2 = \left(\frac{b}{2}\right)^2 - c.$$

Example: Solve by completing the square.

$$x^2 + 6x - 7 = 0$$

Solution:

Move the constant:

$$x^2 + 6x = 7$$

Add $\left(\frac{6}{2}\right)^2 = 9$ to both sides:

$$x^2 + 6x + 9 = 16$$

Now the left side is a perfect square:

$$(x + 3)^2 = 16$$

Take the square root:

$$x + 3 = \pm 4$$

Subtract 3:

$$x = -3 \pm 4$$

Solution set: $\{-7, 1\}$.

Definition

Deriving the Quadratic Formula (from completing the square):

Starting with the general form:

$$ax^2 + bx + c = 0.$$

Divide by a (since $a \neq 0$):

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Move the constant:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

Add $\left(\frac{b}{2a}\right)^2$ to both sides:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Now the left side is a perfect square:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Take the square root:

$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4ac}}{2a}.$$

Subtract $\frac{b}{2a}$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Example: Solve using the quadratic formula.

$$2x^2 - 3x - 5 = 0$$

Solution:

Here $a = 2$, $b = -3$, $c = -5$.

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-5)}}{2(2)} = \frac{3 \pm \sqrt{9 + 40}}{4} = \frac{3 \pm 7}{4}.$$

$$\text{Solution set: } \left\{ \frac{5}{2}, -1 \right\}.$$

Definition

The Discriminant:

In the quadratic formula, the expression $D = b^2 - 4ac$ is called the **discriminant**.

$$D > 0 \Rightarrow \text{Two distinct real solutions.}$$

$$D = 0 \Rightarrow \text{One real (repeated) solution.}$$

$$D < 0 \Rightarrow \text{Two complex (non-real) solutions.}$$

Example: Determine the type of solutions.

$$x^2 + 4x + 5 = 0$$

Solution:

$$D = b^2 - 4ac = 16 - 20 = -4.$$

Since $D < 0$, the equation has two complex solutions.

Real-World Example: Projectile Motion

The height of a ball thrown upward from the ground is modeled by

$$h(t) = -16t^2 + 64t,$$

where h is in feet and t is in seconds. Find how long it takes for the ball to hit the ground.

Solution:

$$h(t) = 0 \Rightarrow -16t^2 + 64t = 0$$

Factor:

$$-16t(t - 4) = 0$$

$$t = 0 \text{ or } t = 4.$$

The ball hits the ground after 4 seconds.

Extra Practice

Solve each equation.

1) $x^2 + 10x + 9 = 0$

2) $3x^2 - 5x - 2 = 0$

3) $x^2 + 4x = 1$

4) $2x^2 + 7x + 3 = 0$

5) $x^2 - 8x + 16 = 0$

6) $x^2 + 6x + 13 = 0$